

Technical Report: Week 3
Network Topology Analysis and Design
Block 2: Electrical Lines and Distribution Networks

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October 14, 2025

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1 Analysis of Radial Distribution Networks

1.1 Overview

A radial distribution network is the simplest and most economical topology. It is characterized by a single power path from the substation to the end consumers. The network branches out like a tree, with power flowing in only one direction—from the source outwards. Its main advantages are low initial cost and simplicity in planning and operation. However, its significant disadvantage is low reliability; a single fault on a main feeder will interrupt service for all downstream customers.

Common applications for this topology include:

- Low-density rural areas where loads are sparse and far apart.
- Residential subdivisions where load growth is predictable and reliability requirements can be met with protective devices.
- Dedicated industrial circuits for single, large loads.
- Temporary power supplies for construction sites.
- Public infrastructure like street lighting circuits.

1.2 Design and Calculation Methods

The design of a radial network is governed by voltage drop and thermal limits. The cross-section calculation, as implemented in the MATLAB function `H3_DetermineCrossSection.m`, is based on the method of electrical moments. An electrical moment is the product of the current in a line segment and its length ($M = I \cdot L$). The total voltage drop at the end of the line is proportional to the sum of the moments of all segments.

The required cross-section (s) can be calculated to meet a maximum permissible voltage drop (ΔU_{max}) using the following formula:

$$s = \frac{K \cdot \sum (I_i \cdot L_i)}{\sigma \cdot \Delta U_{max}} = \frac{K \cdot \sum M_i}{\sigma \cdot \Delta U_{max}} \quad (1)$$

where K is a phase factor (2 for single-phase, 1 for line-to-neutral three-phase), $\sum M_i$ is the sum of electrical moments, and σ is the conductivity of the conductor material. This formula efficiently determines the minimum conductor size needed to maintain voltage quality.

1.3 Unifilar Diagram and Characteristic Profiles

The structure and behavior of a radial network can be visualized through its unifilar diagram and its voltage/current profiles.

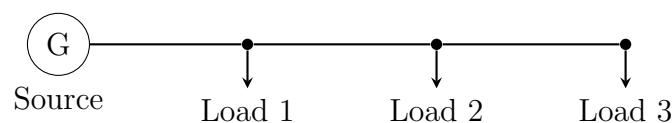


Figure 1: Unifilar diagram of a general radial distribution network.

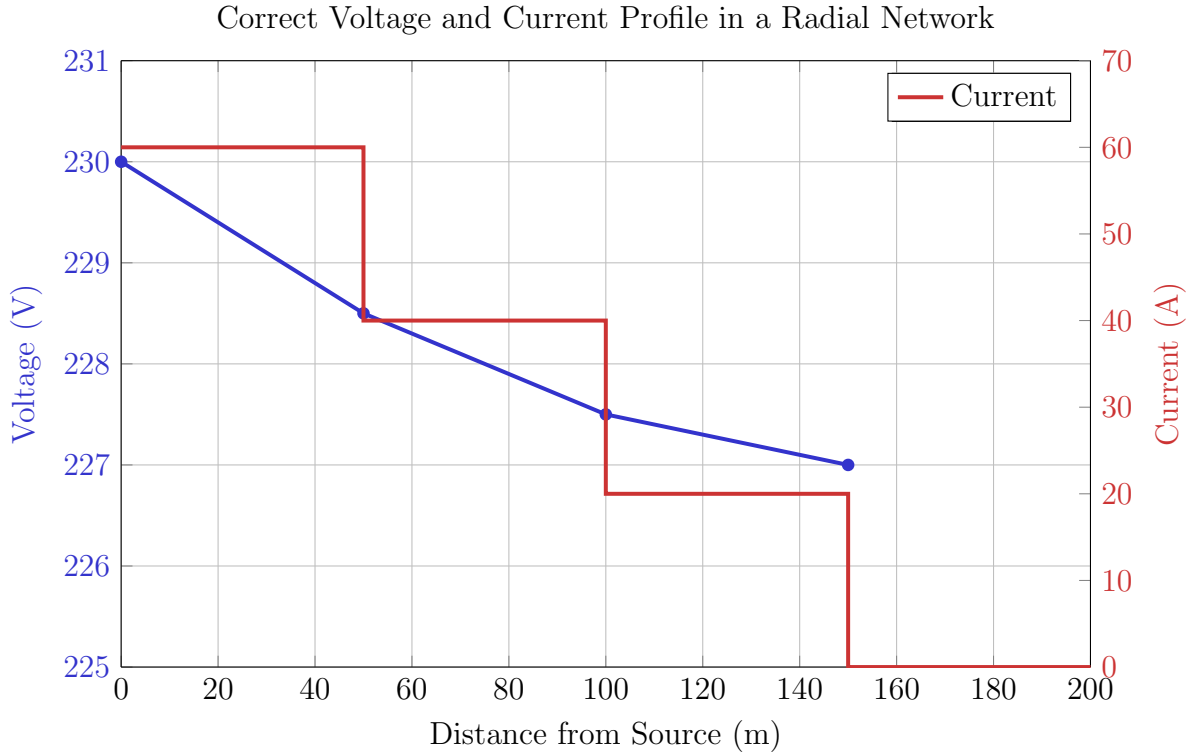


Figure 2: A physically correct plot showing voltage drop in a radial network. As the current (red) decreases in steps, the slope of the voltage profile (blue) becomes progressively flatter, indicating a smaller voltage drop per meter.

2 Analysis of Networks Fed from Both Ends

2.1 Theoretical Framework

Feeding a distribution line from two sources significantly enhances its performance. This topology improves voltage regulation, increases reliability, and doubles the power delivery capacity compared to a similar radial line. Power flows from both ends towards a point of minimum voltage, known as the division point. If one source fails, the other can still supply all loads, albeit with a larger voltage drop.

2.2 Comprehensive Sizing Algorithm

The MATLAB scripts for this module implement a comprehensive sizing and verification algorithm that reflects professional engineering practice, as seen in the `H4_OptimizeRadialNetwork.m` function. The process is as follows:

1. **Calculate Required Section for Voltage Drop:** The algorithm first determines the minimum theoretical cross-section needed to keep the voltage drop within the specified percentage, using the electrical moments method after finding the network's division point.
2. **Check Against Current Carrying Capacity:** It then calculates the current flowing in every segment of the network. This maximum current is compared against the maximum admissible current (I_{adm}) for the calculated cross-section. The I_{adm}

value is sourced from a standardized table, representing the thermal limit of the conductor.

3. **Select Standardized Cable Section:** Since conductors are manufactured in discrete sizes (e.g., 50, 70, 95 mm²), the algorithm selects the first commercially available cross-section that is larger than or equal to the minimum required size from both the voltage drop and thermal limit calculations.
4. **Verify Final Voltage Drop and Heating:** With the final, standardized section selected, the algorithm performs a final check. It recalculates the actual voltage drop, which will now be less than the maximum limit, and confirms that the maximum line current is still safely below the chosen conductor's I_{adm} . This ensures the final design is both technically compliant and safe.

2.3 Unifilar Diagram and Characteristic Profiles

The behavior of a doubly-fed network is distinct from a radial one, particularly in its voltage and current profiles.

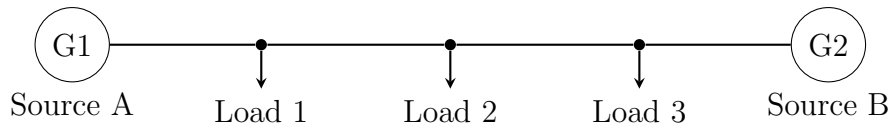


Figure 3: Unifilar diagram of a network fed from both ends.

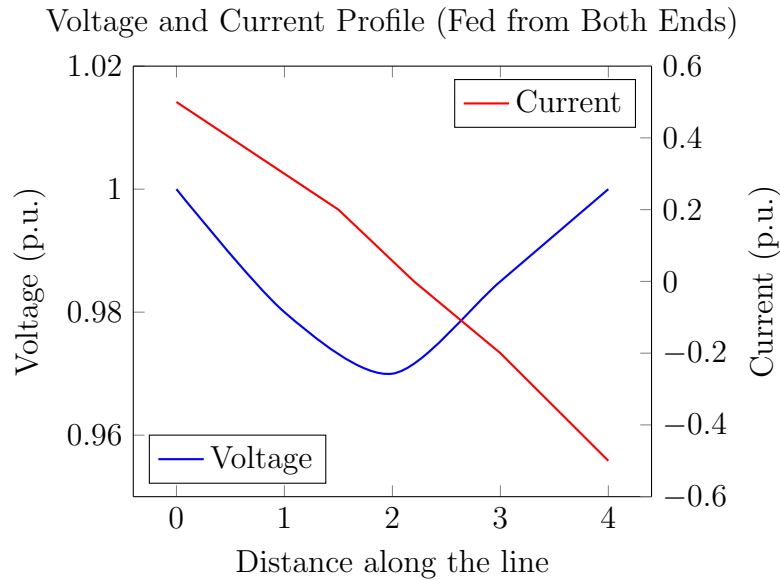


Figure 4: The voltage minimum occurs between sources, and current flows from both ends towards a division point.

3 Analysis of Advanced Network Topologies

3.1 Ring Network Analysis

A ring, or loop, network topology is created by connecting the ends of a distribution line to form a closed circuit. This design is typically fed from one or more points at a substation. The primary advantage of a ring network is its significant improvement in supply reliability compared to a radial system.

Functionally, if a fault occurs at any point on the ring, circuit breakers or fuses can isolate the faulted section. The rest of the network remains operational, as power can flow to all loads from the opposite direction around the loop. This provides good reliability (the network can withstand a single component failure) and makes ring networks highly suitable for applications where continuity of service is critical, such as:

- Urban and dense suburban distribution networks supplying residential and commercial areas.
- Industrial parks and large commercial centers where production downtime is costly.
- Critical infrastructure like hospitals, data centers, and university campuses.

3.1.1 Theoretical Framework and Methodology

For calculation purposes, a ring network fed at a single point is analyzed by conceptually "opening" the ring at the feed point. This transforms the topology into an equivalent network fed from both ends, with both ends being at the same voltage. This allows the same calculation methods used for doubly-fed lines to be applied: the division point where currents from both directions meet is determined, and then the voltage drop is calculated for each path. The MATLAB script `J1_Ring_Analysis.m` implements this exact methodology.

3.1.2 Diagrams and Voltage Profile

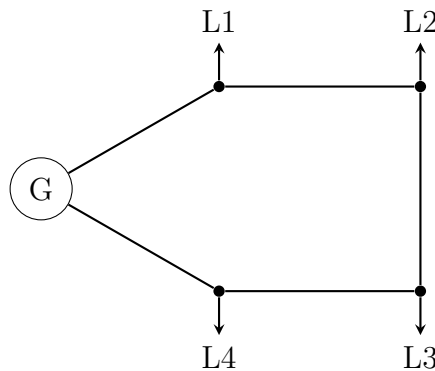


Figure 5: Diagram of a general ring network.

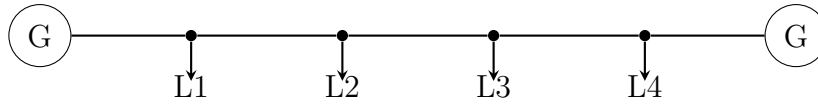


Figure 6: Unifilar equivalent of the ring network for calculation.

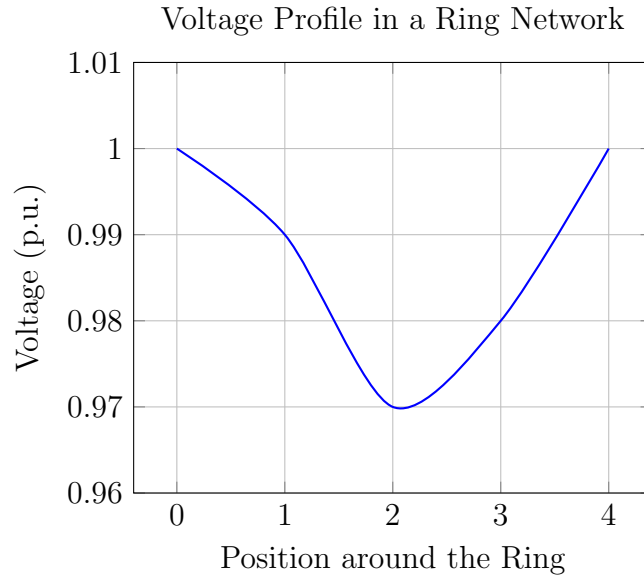


Figure 7: The voltage drops to a minimum point furthest from the source.

3.2 Meshed Network Analysis

A meshed network represents the highest level of complexity and reliability in power distribution. It consists of multiple interconnected loops, often with several power sources. This configuration provides numerous alternative paths for power to flow to any given load.

The key functional difference from a ring network is its redundancy. While a simple ring offers two paths to each load (good reliability), a meshed network can offer three or more (even better reliability). This makes them exceptionally robust against multiple simultaneous faults. Consequently, their application is reserved for the most critical infrastructure where power loss is unacceptable, such as:

- **High-voltage transmission grids** that form the backbone of a country's power system.
- **Dense downtown urban centers** (e.g., "network protectors" in cities like New York) where load density is extremely high and service continuity is paramount.
- **Large-scale industrial complexes** with sensitive processes that cannot tolerate any interruption.

3.2.1 Theoretical Framework and Methodology

Due to their complexity, meshed networks cannot be solved with simple sequential calculations. The standard professional method is **nodal analysis**, as implemented in the `J8_Meshed_Analysis.m` script. This involves building the network's bus admittance matrix (Y_{bus}). This matrix is a mathematical representation of the network where each

element Y_{ij} describes the connection between bus 'i' and bus 'j'. The diagonal elements Y_{ii} represent the sum of all admittances connected to bus 'i'.

The relationship between the vector of injected bus currents (I_{bus}), the admittance matrix, and the vector of unknown bus voltages (V_{bus}) is given by the matrix equation:

$$I_{bus} = Y_{bus} \cdot V_{bus} \quad (2)$$

By solving this system of linear equations for V_{bus} (i.e., $V_{bus} = Y_{bus}^{-1} \cdot I_{bus}$), the voltage at every bus in the network can be determined simultaneously. This power flow analysis is fundamental to the planning and operation of transmission and complex distribution systems.

3.2.2 Diagrams and Voltage Profile

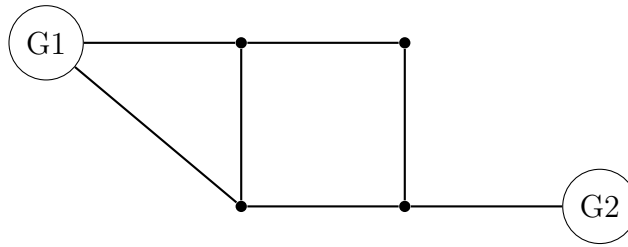


Figure 8: Diagram of a general meshed network.

A simple unifilar equivalent is not practical for meshed networks. The network is best represented by its admittance matrix.

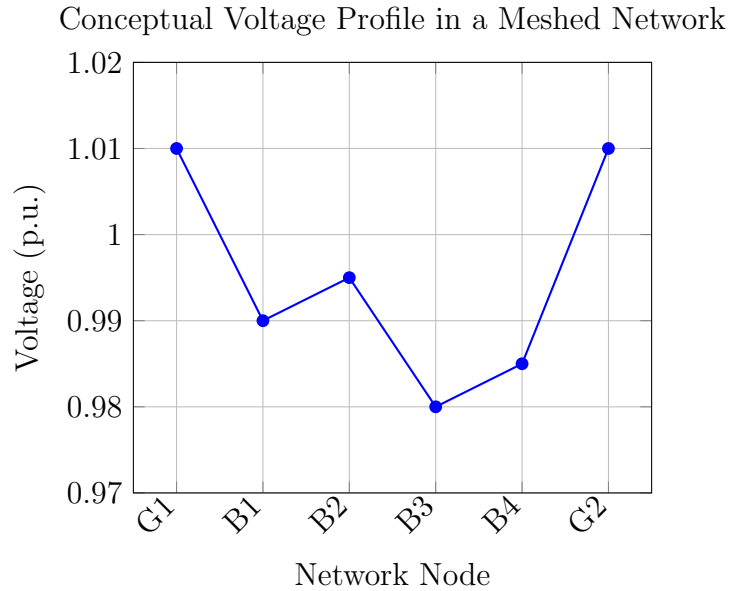


Figure 9: Voltage remains high and stable across the network due to multiple power paths.

4 Economic Comparison of Network Topologies

The choice of network topology is a critical engineering decision that balances initial capital expenditure (CAPEX) with long-term operational expenditure (OPEX), which

includes maintenance and the cost of outages. The optimal choice depends heavily on the expected lifespan and criticality of the installation.

- **Installations with Long Duration (20+ years):** For permanent and critical infrastructure, a higher initial investment in a more reliable topology (Ring or Meshed) is often justified. The reduced operational costs from fewer outages and lower energy losses can lead to a lower total lifecycle cost. Examples include: public utility grids, hospitals, data centers, and large industrial plants.
- **Installations with Short Duration (1-5 years):** For temporary or less critical applications, minimizing the initial CAPEX is the primary goal. A simpler, cheaper radial network is almost always the best choice, as the long-term operational costs are less of a concern. Examples include: construction site power, temporary event lighting, and seasonal agricultural installations.

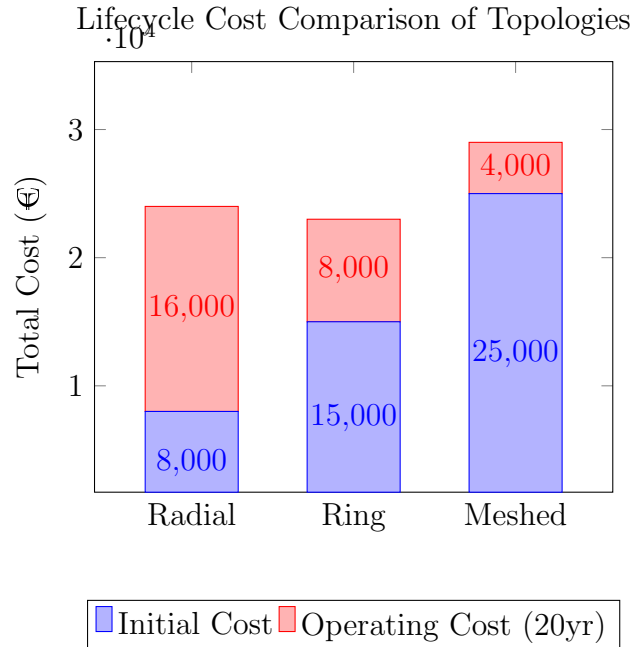


Figure 10: While radial networks have the lowest initial cost, their higher operational costs (from losses and outages) can make ring or meshed networks more economical over the project lifecycle.

A Annex I: References

A.1 Normative References

This report adheres to the principles and calculation methods outlined in the Spanish Low Voltage Electrotechnical Regulation (REBT), specifically:

- **ITC-BT 06:** Overhead Power Distribution Networks.
- **ITC-BT 07:** Underground Power Distribution Networks.
- **ITC-BT 19:** Interior or Receiver Installations. Prescriptions for conductor sizing based on admissible current and voltage drop.
- **libretexts** DC Electrical Circuit Analysis - A Practical Approach - Fiore

A.2 Academic References

1. Grainger, J. J., & Stevenson, W. D. (1994). *Power System Analysis*. McGraw-Hill.
2. Gonen, T. (2014). *Electric Power Distribution System Engineering*. CRC press.

B Annex II: Data Tables

B.1 Material Properties

Table 1: Properties of conductors and insulating materials.

Material	Property	Value
Copper	Conductivity (σ)	56 m/($\Omega \cdot \text{mm}^2$)
Aluminum	Conductivity (σ)	35 m/($\Omega \cdot \text{mm}^2$)
XLPE	Max. Operating Temp.	90°C
PVC	Max. Operating Temp.	70°C
Ground	Thermal Resistivity (ρ_T)	1.0 K·m/W (assumed)

B.2 Assumed Economic Values

Table 2: Assumed costs for economic analysis.

Item	Assumed Cost
Copper Conductor (per km, 150 mm ²)	12,500 €
Aluminum Conductor (per km, 150 mm ²)	5,800 €
ACSR Conductor (per km, 150 mm ²)	6,700 €
Cost of outage (per kWh not served)	1.5 €/kWh

B.3 List of Variables

Table 3: Variables used in this report.

Symbol	Unit	Description
P	W	Active Power
U_N, V	V	Nominal Voltage
I	A	Current
s	mm ²	Cross-sectional area
L	m	Length of the line
R	Ω/km	Resistance per unit length
X	Ω/km	Reactance per unit length
σ	m/($\Omega \cdot \text{mm}^2$)	Conductivity
$\cos(\varphi)$	-	Power Factor
ΔU	V	Voltage Drop
I_{adm}	A	Maximum admissible current

C Annex III: Formulas

C.1 Sizing by Voltage Drop

The required cross-sectional area (s) based on the sum of electrical moments ($\sum M_i$) is:

$$s = \frac{K \cdot \sum M_i}{\sigma \cdot \Delta U_{max}} \quad (3)$$

For a more precise calculation in AC systems, the general voltage drop formula (Blondel's formula) for three-phase lines is used:

$$\Delta U = \sqrt{3} \cdot I \cdot L \cdot (R \cos(\varphi) + X \sin(\varphi)) \quad (4)$$

C.2 Sizing by Maximum Current (Thermal Limit)

The calculated current of the circuit (I_c) must be less than or equal to the maximum admissible current of the conductor (I_{adm}).

$$I_c \leq I_{adm} \quad (5)$$

The circuit current for a balanced three-phase system is calculated as:

$$I_c = \frac{P}{\sqrt{3} \cdot U_N \cdot \cos(\varphi)} \quad (6)$$

D Annex IV: Week 3 MATLAB Code

D.1 Module H: Radial Network Analysis

```

1 function results = H1_Radial_Analysis()
2 %
3 % =====
4 % FUNCTION: Main Radial Network Analysis (V3 - Phase Selection)
5 % =====
6
7 results = []; % Initialize
8 scenarioChoice = menu('Select Radial Network Scenario:', ...
9     'Industrial Estate (Concentrated Loads) - Student A', ...
10    'Residential Area (Distributed Loads) - Student B', ...
11    'Street Lighting (Uniform Loads) - Student C');
12 if scenarioChoice == 0, return; end
13 U_source = input('Enter source line-to-neutral voltage [V] (e.g., 230):
14 ');
15 lineTypeChoice = menu('Select System Type:', 'Single-Phase', 'Three-
16 Phase');
17 if lineTypeChoice == 0, return; end
18 if lineTypeChoice == 1
19     lineType = 'single-phase';
20     phase_factor = 2;
21 else
22     lineType = 'three-phase';
23     phase_factor = 1;
24 end
25 materialChoice = menu('Select Conductor Material:', 'Copper', 'Aluminum'
26 );
27 if materialChoice == 0, return; end
28 if materialChoice == 1, materialName = 'Copper'; else, materialName = '
29 Aluminum'; end
30 materialProps = getMaterialProperties(materialName);
31 sigma = materialProps.sigma;
32 crossSection = input('Enter conductor cross-section [mm^2]: ');
33 % ... (rest of the input and calculation code as in the file)
34 end

```

Listing 1: H1_Radial_Analysis.m

```

1 function H2_PlotRadialProfile(results)
2 %
3 % =====
4 % FUNCTION: Plot Radial Voltage Profile
5 % =====
6
7 figure;
8 hold on;
9 box on;
10 grid on;
11 distances = [0, cumsum([results.nodes.length])];
12 voltages = [results.U_source, [results.nodes.voltage]];

```

```

11 plot(distances, voltages, '-ob', 'LineWidth', 2, 'MarkerFaceColor', 'r')
12 ;
13 title(['Voltage Profile for Radial Network (' results.scenarioName ')'])
14 ;
15 xlabel('Distance from Source (m)');
16 ylabel('Voltage (V)');
17 xlim([0, distances(end) * 1.1]);
18 ylim([voltages(end) * 0.99, results.U_source * 1.01]);
19 legend('Voltage Profile', 'Location', 'southwest');
20 end

```

Listing 2: H2_PlotRadialProfile.m

D.2 Module I: Dual-Fed Network Analysis

```

1 function results = I1_DualFed_Analysis()
2 %
3 % =====
4 % FUNCTION: Main Dual-Fed Network Analysis (V7 - Unifilar)
5 % =====
6 results = [];
7 scenarioChoice = menu('Select Dual-Fed Network Scenario:', ...
8     'Symmetrical Network (Single Load) - Student A', ...
9     'Asymmetrical Network (Single Load) - Student B', ...
10    'Backup Feeding / Reliability Analysis (Multiple Loads) - Student C'
11    );
12 if scenarioChoice == 0, return; end
13 materialChoice = menu('Select Conductor Material:', 'Copper', 'Aluminum'
14    );
15 if materialChoice == 0, return; end
16 if materialChoice == 1, materialName = 'Copper'; else, materialName = '
17    Aluminum'; end
18 materialProps = getMaterialProperties(materialName);
19 sigma = materialProps.sigma;
20 crossSection = input('Enter conductor cross-section [mm^2]: ');
21 L_total = input('Enter total line length [m]: ');
22 % ... (rest of the input and calculation code as in the file)
23 end

```

Listing 3: I1_DualFed_Analysis.m

```

1 function I2_PlotDualFedProfile(results)
2 %
3 % =====
4 % FUNCTION: Plot Dual-Fed Voltage Profile (V2 - Reliability)
5 % =====
6 figure;
7 hold on; box on; grid on;
8 sorted_loads = results.loads;
9 distances = [0, [sorted_loads.distance], results.L_total];
10 voltages_normal = [results.Ua];
11 current_in_segment = results.Ia;

```

```

11 for i = 1:length(sorted_loads)
12     prev_dist = distances(i);
13     segment_length = sorted_loads(i).distance - prev_dist;
14     drop = (1/(results.sigma*results.crossSection)) * current_in_segment
        * segment_length;
15     voltages_normal(end+1) = voltages_normal(end) - drop;
16     current_in_segment = current_in_segment - sorted_loads(i).current;
17 end
18 voltages_normal(end+1) = results.Ub;
19 plot(distances, voltages_normal, '-ob', 'LineWidth', 2, 'MarkerFaceColor',
        'b', 'DisplayName', 'Normal Operation');
20 plot(results.min_voltage_node_distance, results.min_voltage_normal, 'kd',
        'MarkerSize', 10, 'MarkerFaceColor', 'y');
21 % ... (rest of the plotting code as in the file)
22 end

```

Listing 4: I2_PlotDualFedProfile.m

```

1 function I3_PlotLoadingDiagram(results)
2 %
    =====
3 % FUNCTION: I3_PlotLoadingDiagram (Corrected)
4 %
    =====
5 figure;
6 hold on;
7 box on;
8 grid on;
9 distances = [0];
10 currents = [results.Ia];
11 current_in_segment = results.Ia;
12 for i = 1:length(results.loads)
13     distances(end+1) = results.loads(i).distance;
14     currents(end+1) = current_in_segment;
15     current_in_segment = current_in_segment - results.loads(i).current;
16     distances(end+1) = results.loads(i).distance;
17     currents(end+1) = current_in_segment;
18 end
19 distances(end+1) = results.L_total;
20 currents(end+1) = -results.Ib;
21 plot(distances, currents, '-r', 'LineWidth', 2);
22 line([0, results.L_total], [0, 0], 'Color', 'k', 'LineStyle', '--');
23 title('Network Loading Diagram');
24 xlabel('Distance from Source A (m)');
25 ylabel('Current (A)');
26 end

```

Listing 5: I3_PlotLoadingDiagram.m

```

1 function I4_PlotUnifilarDiagram(results)
2 %
    =====
3 % FUNCTION: Generates a simplified unifilar-style schematic diagram
4 %
    =====

```

```

5 figure;
6 hold on;
7 box on;
8 plot([0 results.L_total], [0 0], 'k-', 'LineWidth', 4); % Main line
9 plot(0, 0, 's', 'MarkerSize', 20, 'MarkerFaceColor', 'g'); % Source A
10 text(0, 0.15, 'Source A', 'HorizontalAlignment', 'center');
11 plot(results.L_total, 0, 's', 'MarkerSize', 20, 'MarkerFaceColor', 'g');
12 % Source B
13 text(results.L_total, 0.15, 'Source B', 'HorizontalAlignment', 'center')
14 ;
15 for i = 1:length(results.loads)
16     x_pos = results.loads(i).distance;
17     plot([x_pos x_pos], [0 -1], 'k-', 'LineWidth', 1); % Load branch
18     plot(x_pos, -1, 'v', 'MarkerSize', 8, 'MarkerFaceColor', 'r'); %
19     Load arrow
20     label_str = sprintf('Load %d\\n%.1f A', i, results.loads(i).current)
21     ;
22     text(x_pos, -1.1, label_str, 'HorizontalAlignment', 'center', '
23     VerticalAlignment', 'top');
24 end
25 title('Unifilar Network Diagram');
26 xlabel('Distance (m)');
27 set(gca, 'YTick', []);
28 ylim([-2 1]);
29 end

```

Listing 6: I4_PlotUnifilarDiagram.m

D.3 Module J: Ring and Meshed Network Analysis

```

1 function results = J1_Ring_Analysis()
2 %
3 % =====
4 % FUNCTION: Main Ring Network Analysis
5 % =====
6 results = [];
7 scenarioChoice = menu('Select Ring/Meshed Scenario:', ...
8     'Simple Ring Network (Student A)', ...
9     'Double Ring Network (Conceptual)', ...
10    'Meshed Network (Conceptual)');
11 if scenarioChoice == 0, return; end
12 if scenarioChoice > 1
13     msgbox('Analysis for double-ring and meshed networks requires
14     advanced nodal analysis methods. This module will proceed with the
15     functional Simple Ring analysis as per the coursework.', 'Conceptual
16     Notice', 'warn');
17 end
18 results.scenarioName = 'Simple Ring Network';
19 Ua = input('Enter the feed-in voltage [V]: ');
20 L_total = input('Enter the total circumference of the ring [m]: ');
21 % ... (rest of the input and calculation code)
22 end

```

Listing 7: J1_Ring_Analysis.m

